

Briefly on Grammatical Representation

What is Grammar?

- Several definitions:
 - The rules and principles that guide the structure of sentences in language
 - Regularities in the mapping between phonological form and semantic form
 - The way a language speaker organizes her experience with language, allowing her to make generalizations

Why do we need grammar?

- Grammatical analysis in NLP involves breaking the text into simpler parts and categorizing them
 - New sentences are the rule, rather than the exception (unlike new words)
- Two main motivations for grammar:
 - **Structural Analysis:** decomposing complex units into simpler sub-structures can assist learning and generalization
 - **Semantic Information:** grammatical structure often reflects semantic structure and distinctions

Text Strings are not Informative Enough, even with POS Tags

Q: Where were the crates taken?

The terrorists took the crates they found **to** the hangar

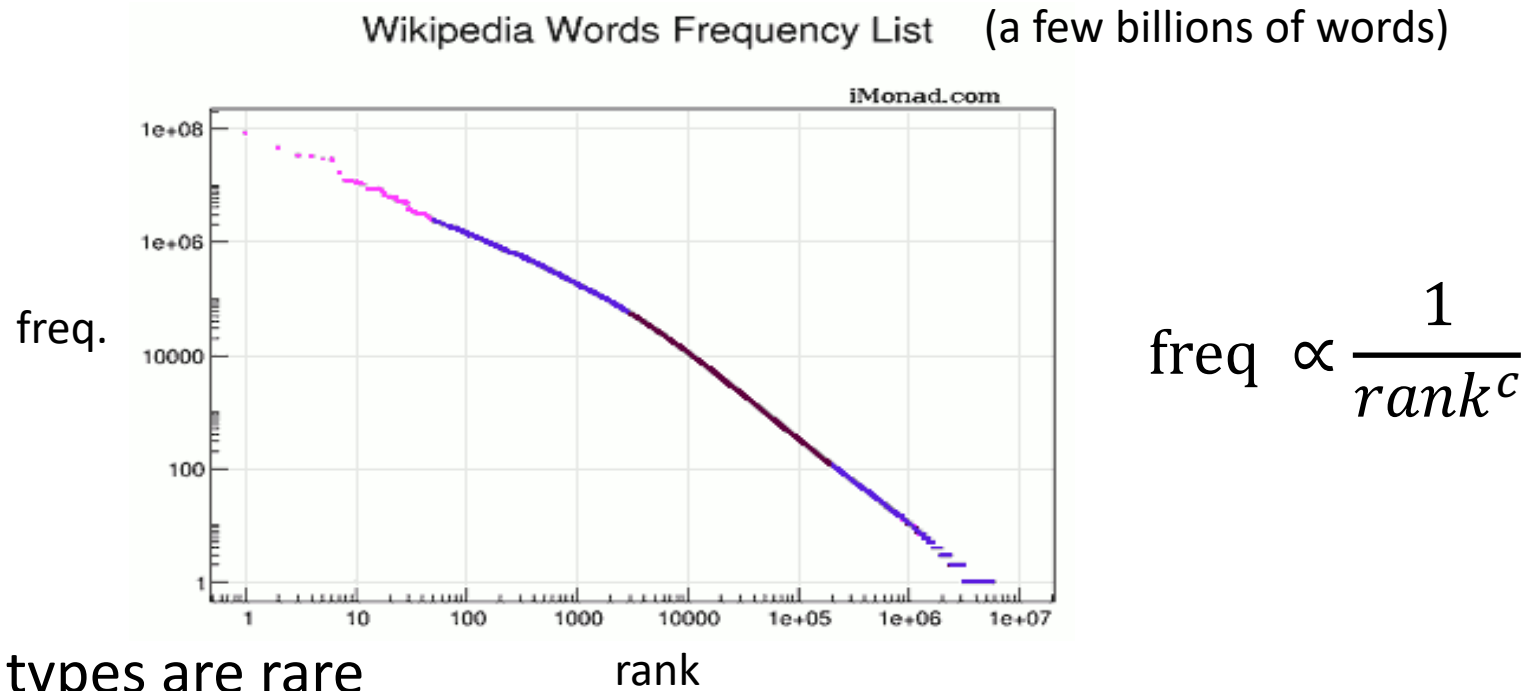
A: to the hangar

The terrorists took the crates they found **in** the hangar

A: *not* to the hangar

The Importance of Categorization

- Reminder: linguistic phenomena tend to have a power-law distribution



- Most word types are rare
- Many word instances are rare

The Importance of Categorization

- This means that for most types, we don't have meaningful statistics
 - Many types don't appear at all
- Categorization allows information to propagate from one type to another

